



A Singleton design pattern ensures that only one instance of a class is created.

Imagine a disorganised wardrobe with heaps of clothes in every section. How would you organise it? When I asked my students this question, I got the following answers:

1. Empty the wardrobe
2. Arrange the clothes
 - on the basis of the most frequently used ones
 - or the favourite ones
 - or by their type (e.g., Indian/Western)
3. Fix each section for each category and arrange the clothes accordingly

If a similar problem occurs in another context, such as when re-arranging a bookshelf, the same solution can be reused. This is the basic principle behind design patterns. With respect to object-oriented software engineering, a design pattern describes a redundant problem and provides a reusable solution.

Types of design patterns

Design patterns have been categorised by the Gang of Four or GoF (Gamma, Helm, Johnson and Vlissides) into three types:

1. Creational – Patterns that deal with object creation
2. Structural – Patterns that deal with the composition of classes or objects, and their relationships
3. Behavioural – Patterns that focus on how objects distribute work, and how each object working independently can help achieve a common goal

The Singleton design pattern

Let us look at the website of a restaurant as an example. The website has a menu link titled *Menu*. Every time a user clicks on this link, data is retrieved from the database and stored in an instance (object) of a class named 'Menu'. The class diagram for 'Menu' is shown in Figure 1.